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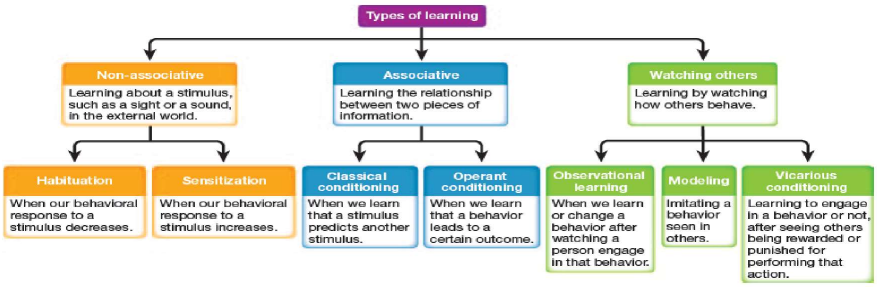
Wednesday, September 18, 2024 9:57 PM

Summary of the book



How Do We Learn?
6.1 Learning Results from Experience
Learning is a relatively enduring change in behavior, resulting from experience. Learning enables animals to better adapt to their environment. The three major types of learning are nonassociative, associative, and social. Nonassociative learning is a change in behavior after repeated exposure to a single stimulus or event. Associative learning is the linking of two stimuli or events. Social learning is acquiring or changing a behavior after being verbally instructed or exposed to another individual performing that behavior.
6.2 Nonassociative Learning Involves Habituation and Sensitization
Habituation leads to a decrease in a behavioral response after repeated exposure to a stimulus. Habituation occurs when the stimulus stops providing new information. Sensitization leads to an increase in a behavioral response after exposure to a repeated stimulus. Sensitization occurs in cases where increased attention to a stimulus may prove beneficial, such as in dangerous or exciting situations.

How Do We Learn Predictive Associations?
6.3 Classical Conditioning Is Learning What Goes Together

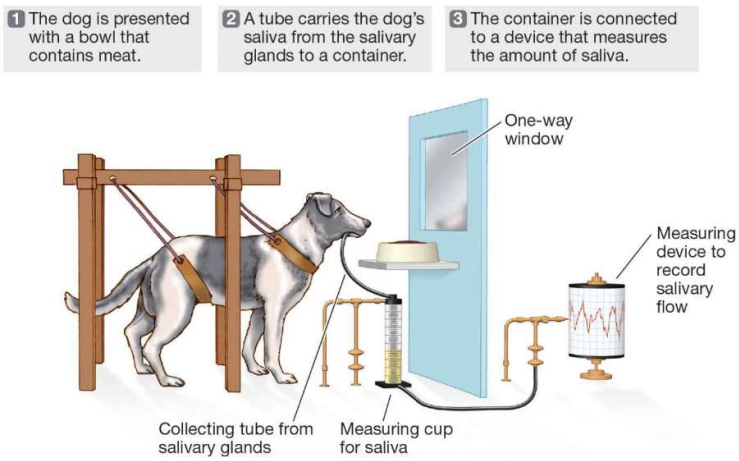
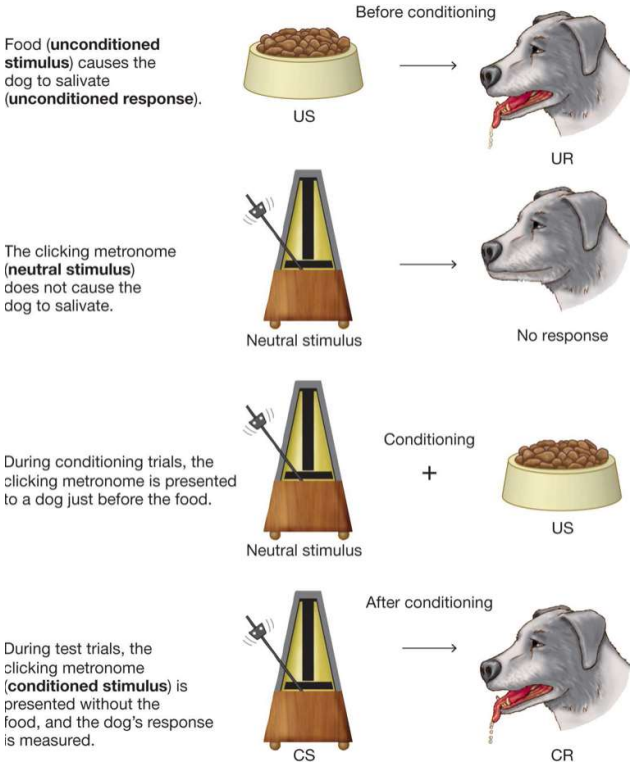
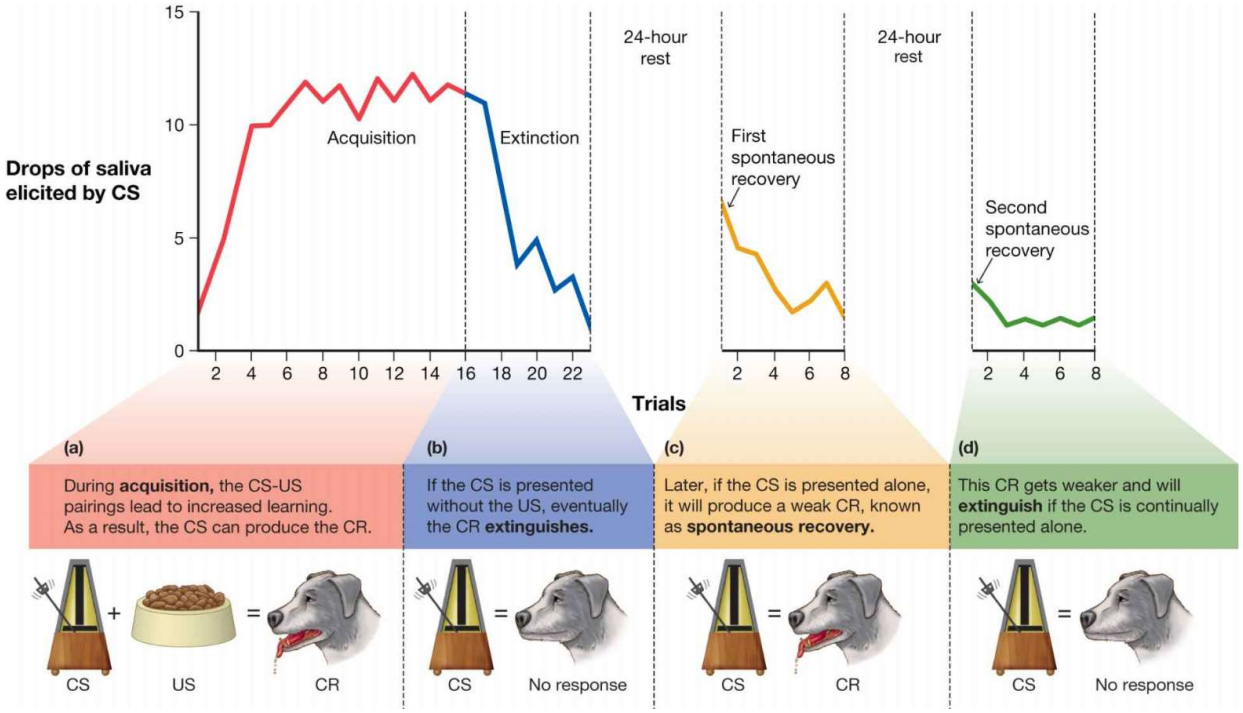


FIGURE 6.6
Pavlov's Apparatus and Classical Conditioning

Pavlov established the principles of classical conditioning. Through classical conditioning, associations are made between two stimuli, such as the clicking of a metronome and the presence of food. What is learned is that one stimulus predicts another.



6.4 Learning Is Acquired and Persists Until Extinction



Learning the association between two stimuli is called acquisition; learning that the association no longer exists when the stimuli are no longer paired is called extinction. Spontaneous recovery occurs when, after a period of time, a previously extinguished conditioned response is once again expressed.

6.5 Learning Involves Expectancies and Prediction

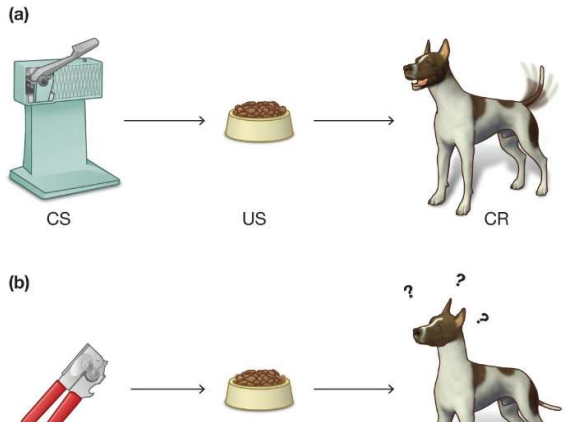
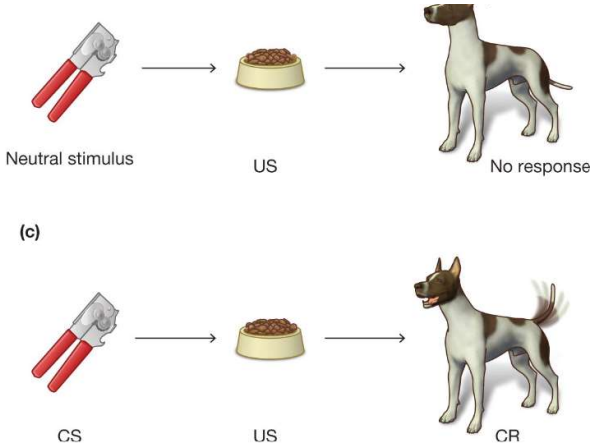


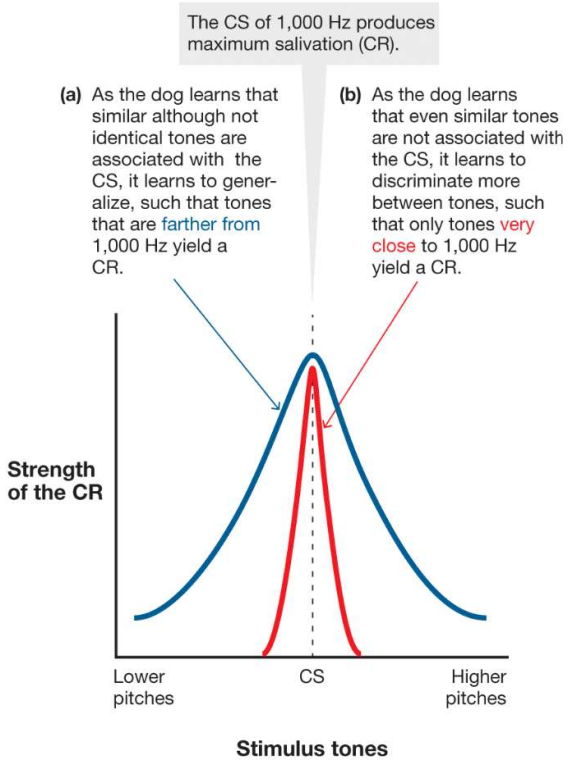
FIGURE 6.10
Rescorla-Wagner Model
The Rescorla-Wagner model of learning emphasizes prediction error.
(a) Here, a dog associates the sound of an electric can opener with the arrival of food.
(b) With the substitution of a manual can opener for the electric one, the dog is initially surprised. What happened to the reliable predictor of the dog's food?
(c) This prediction error causes the dog to check the environment for a new stimulus. When the dog comes to associate the manual can opener with the



of food.
(b) With the substitution of a manual can opener for the electric one, the dog is initially surprised. What happened to the reliable predictor of the dog's food?
(c) This prediction error causes the dog to check the environment for a new stimulus. When the dog comes to associate the manual can opener with the arrival of food, the new stimulus has become the better predictor of the expected event: time to eat!

The Rescorla-Wagner theory describes how the strength of association between two stimuli depends on how unexpected or surprising the unconditioned stimulus is. Positive prediction error results when an unexpected stimulus is presented. Positive prediction error strengthens the association between the CS and the US. Negative prediction error results when an expected stimulus is missing. Negative prediction error weakens the CS-US association.

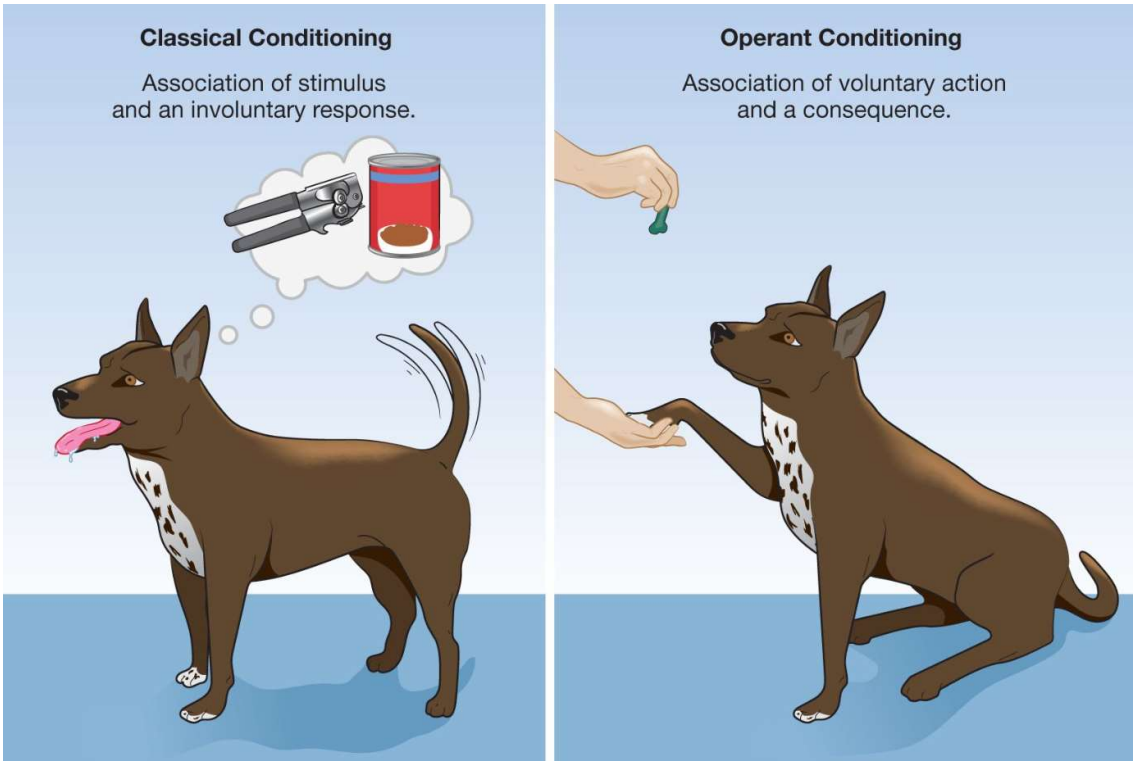
6.6 Learning Shapes Both Conditioned and Unconditioned Stimuli



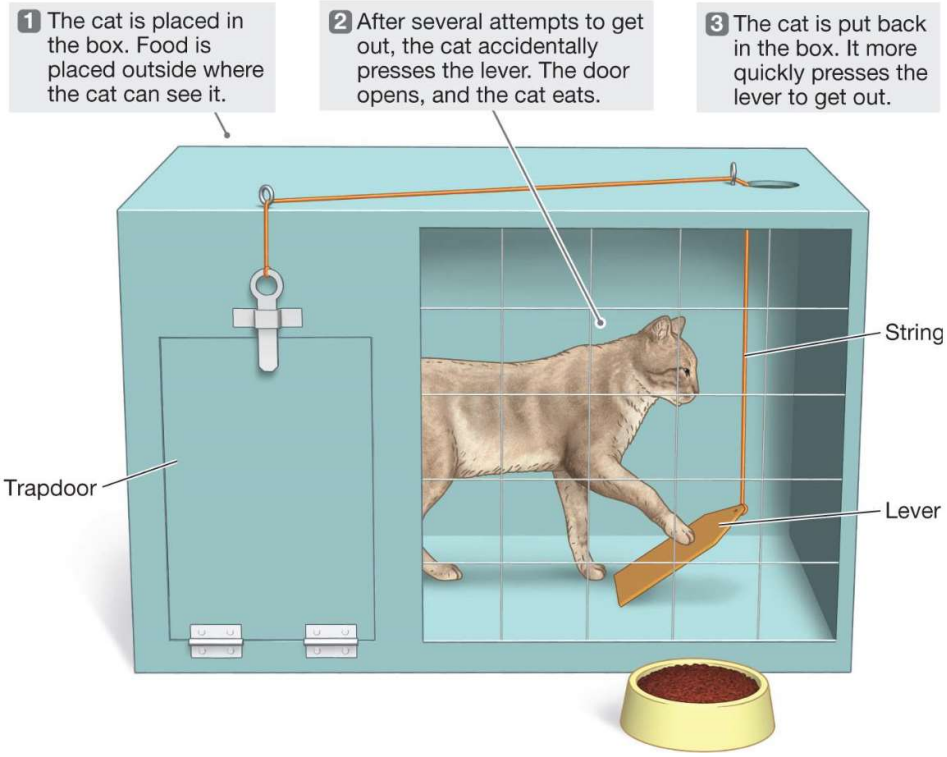
Once an association between a CS and US has become so well-established that the CS acquires value, other stimuli that are associated with the CS alone may also become conditioned through second-order conditioning. Stimuli that are similar to a conditioned stimulus may elicit a conditioned response through generalization, but animals can also learn to discriminate between similar stimuli.

How Do Consequences of an Action Shape Behavior?

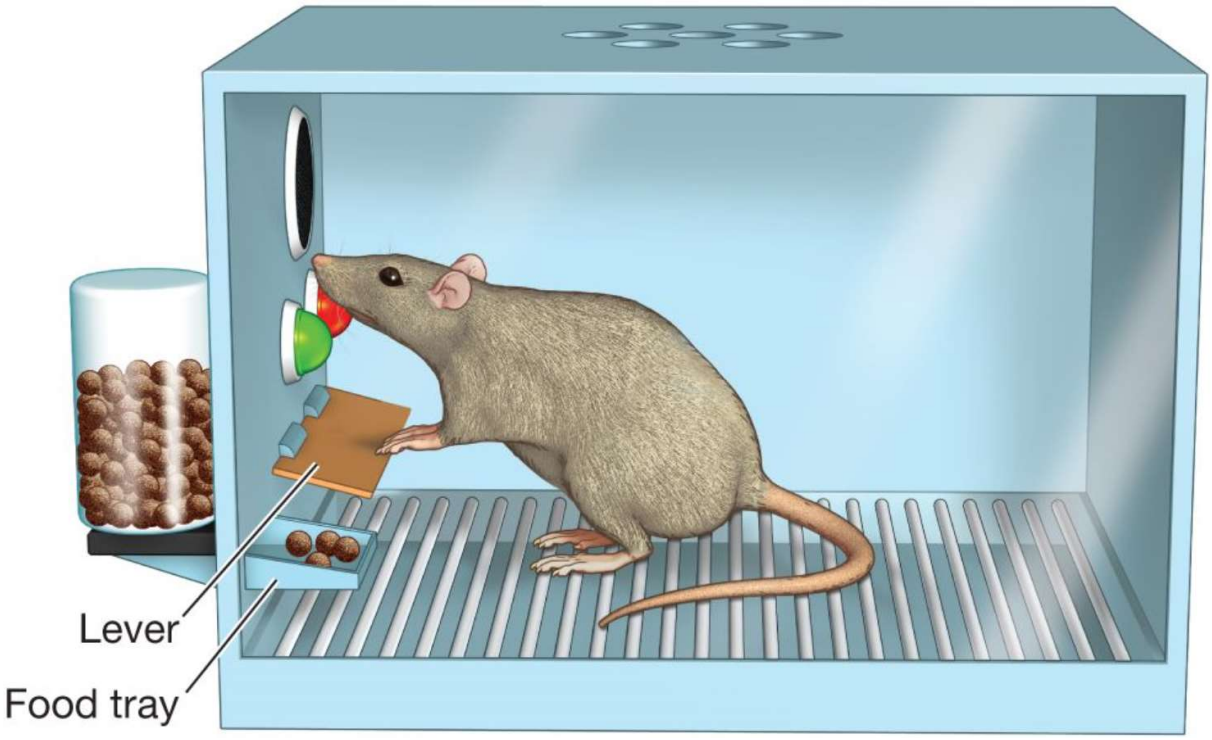
6.7 Operant Conditioning Is Learning Actions from Consequences



Operant conditioning is the learning process in which an action's consequences determine the likelihood of that action being repeated. Through operant conditioning, an animal learns predictive associations between actions and consequences.



6.8 Reinforcement Increases Behavior, Punishment Decreases Behavior



Reinforcement increases the likelihood that a behavior will be repeated. In positive reinforcement, a pleasurable stimulus is delivered after a behavior (e.g., giving a dog a treat for sitting). In negative reinforcement, an aversive stimulus is removed after a behavior (e.g., letting a puppy out of its crate when it acts calmly). Punishment decreases the probability that a behavior will repeat. Positive punishment involves the administration of an aversive stimulus, such as a squirt of water in the face, to decrease behavior. Negative punishment involves the removal of a rewarding stimulus, such as money, to decrease behavior.

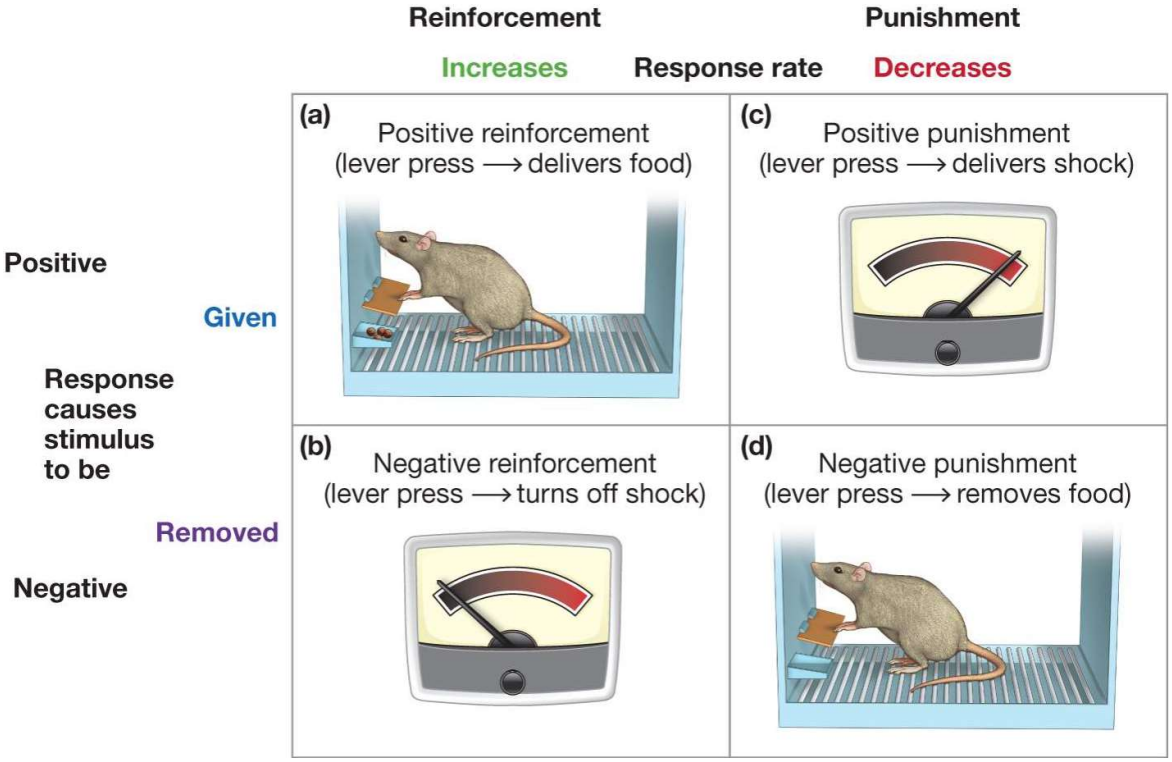
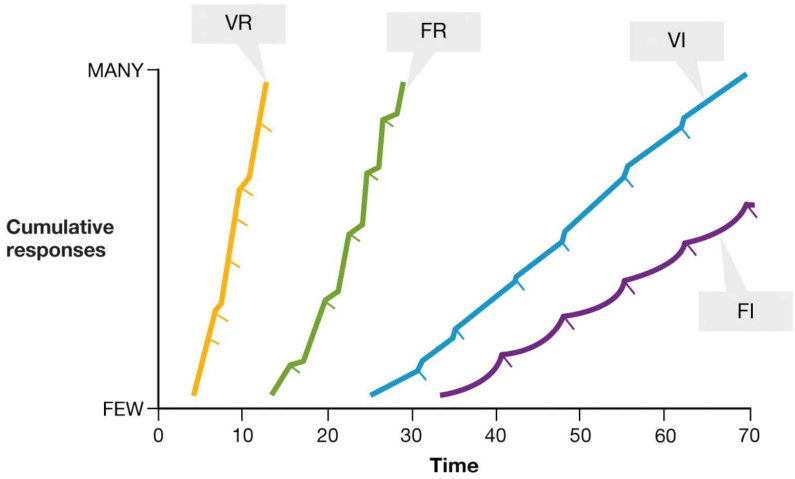


FIGURE 6.17
Positive Reinforcement and Negative Reinforcement, Positive Punishment and Negative Punishment
(a) In positive reinforcement, the response rate increases because responding causes the stimulus to be given.
(b) In negative reinforcement, the response rate increases because responding causes the stimulus to be removed.
(c) In positive punishment, the response rate decreases because responding causes the stimulus to be given.
(d) In negative punishment, the response rate decreases because responding causes the stimulus to be removed.

6.9 Psychology Outside the Lab: When Is Parental Punishment Effective?
Although some forms of parental punishment, such as time-outs, can be effective, spanking is generally ineffective and can lead to a number of negative outcomes. Many psychologists believe that any method of parental punishment is less effective than providing positive reinforcement for "better" behaviors.
6.10 Learning Shapes Actions and Reinforcers
Operant conditioning requires an animal to perform an action to be reinforced. When the desired action is not performed spontaneously, shaping can be used to reinforce successive approximations of the desired behavior. Reinforcers may be primary (those that satisfy biological needs) or secondary (those that do not directly satisfy biological needs but acquire their value through conditioning).
6.11 You Be the Psychologist: Can You Challenge Superstitious Behaviors?
Accidental reinforcement of behaviors occurring close in time can lead to superstitious behaviors in humans and other animals. Although these are generally harmless, at times superstitions can interfere with daily life. Given the human tendency to see associations and believe that they are true, proving that a behavior is superstitious can be challenging.
6.12 Operant Conditioning Is Influenced by Value and Timing
Whether or not an association is learned between an action and consequences can depend on how closely in time they occur. If there is a delay in the consequence of an action, learning can be difficult. Humans can conceptually link an earlier action with a later consequence, but the value of reinforcement may diminish when the consequence occurs in the future. Some reinforcers are more valuable than others, which can influence the likelihood or ease of learning.
6.13 Operant Conditioning Is Influenced by Schedules of Reinforcement



Learning occurs in response to continuous reinforcement and partial reinforcement. Partial reinforcement may be delivered on a ratio schedule or an interval schedule. Each type of schedule may be fixed or variable. Partial reinforcement administered on a variable ratio schedule is particularly resistant to extinction.

Learning Tip

To remember the four schedules of reinforcement:
If the schedule is predictable, it is fixed.
If the schedule is not predictable, it is variable.
If the schedule depends on the passage of time, it is interval.
If the schedule depends on the number of responses, it is ratio.
Put them together:
fixed interval: Reinforcement is given after a predictable passage of time.
variable interval: Reinforcement is given after an unpredictable passage of time.
fixed ratio: Reinforcement is given after a predictable number of responses.
variable ratio: Reinforcement is given after an unpredictable number of responses.

What Do Different Types of Associative Learning Have in Common?

6.14 Learning Is Influenced by Biology and Evolution

An animal's innate biological responses and actions influence what is learned or how easily the animal can learn. For instance, sickness is more likely to be associated with the taste of food eaten hours earlier than another stimulus experienced closer in time. Animals are also predisposed to learn species-typical reward or defensive behaviors from reinforcement or punishment. Stimuli that were natural threats through evolution, such as snakes, are more readily associated with aversive events.

6.15 Dopamine Activity Underlies Learning from Rewards

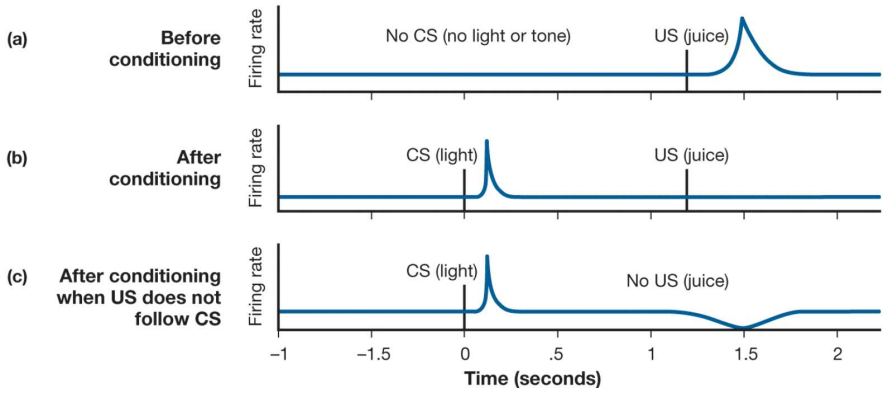


FIGURE 6.26
Prediction Error and Dopamine Activity
Dopamine activity in the brain signals the receipt of a reward.
(a) The blue line clearly shows a spike in dopamine activity. This activity resulted from a positive prediction error after the unexpected arrival of the US.
(b) Once the US was associated with the CS, the spike in dopamine activity occurred after the arrival of the CS but not after the arrival of the expected US.
(c) Dopamine activity continued after the arrival of the CS. However, once the US no longer appeared, negative prediction error resulted in decreased dopamine activity.

The brain has specialized centers that produce pleasure when stimulated. Behaviors and stimuli that activate these centers are reinforced. The nucleus accumbens has dopamine receptors, which are activated by rewarding stimuli. Through conditioning, secondary reinforcers can also activate dopamine receptors. The neurotransmitter dopamine provides one neurobiological basis for learning from prediction errors. Dopamine release increases after positive prediction error and decreases after negative prediction error.

6.16 Phobias and Addictions Have Learned Components

Phobias involve learned fear associations. Addictions involve learned reward associations. Classical and operant conditioning can explain how people learn to associate not only the fearful stimulus or drug itself with the fear or reward but also a host of other "neutral" stimuli. In the case of drug addiction, addicts often inadvertently associate environmental aspects of the purchase and use of the drug with the rewarding consequences and biological changes produced by the drug. These learned associations are major factors in relapse, as seemingly innocuous stimuli can trigger cravings even years after drug use is discontinued.

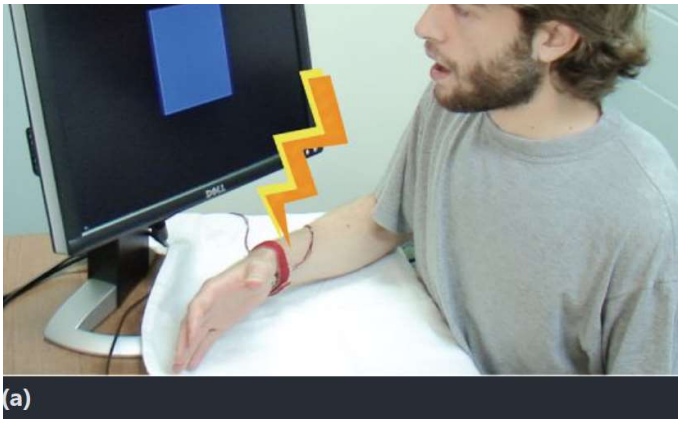
How Do We Learn from Others?

6.17 Social Learning Occurs Through Observation and Instruction

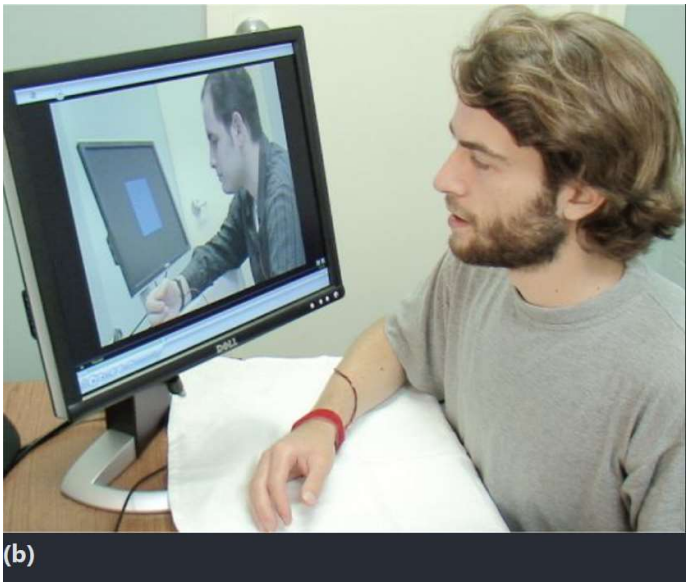
Social learning is a powerful adaptive tool. Humans and other animals learn from being instructed and by watching the behavior of others. The imitation of observed behavior is referred to as modeling. Instructed learning involves acquiring associations and behavior through verbal communication. Vicarious learning occurs when people learn about an action's consequences by observing others being reinforced or punished for their behavior.

6.18 Fear Can Be Learned Through Social Means

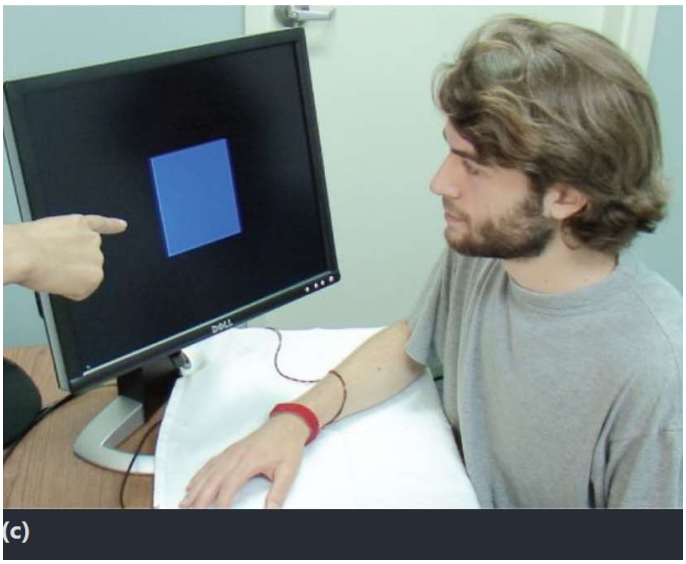
FIGURE 6.35
Social Learning of Fear
Fear can be learned by:
(a) classical conditioning, such as pairing a shape with a mild shock;



conditioning, such as pairing a shape with a mild shock;



(b) observational learning, watching someone else receive a mild shock paired with a shape; or



(c) instructed learning, being told a mild shock will be paired with a shape.

Monkeys have learned to fear snakes (but not flowers) by watching other monkeys react fearfully. These findings suggest that monkeys can learn by observation if the behavior is biologically adaptive. People also learn fear by observation or by instruction. People being instructed that a stimulus is paired with a shock, or observing other people receive a shock, experience activation in the amygdala—a brain area important for processing threatening stimuli—even though they themselves did not receive any shocks.

6.19 Psychology Outside the Lab: Social Learning Has Implications for Criminal Justice

Social learning of cultural associations between Black race and criminality is proposed to contribute to racial bias in the U.S. legal system. The psychologist Jennifer Eberhardt has devoted her career to understanding how race influences policing and legal decisions.